

Shrey Singh

+91 7406631225 | gshrey@gmail.com | gshrey.vercel.app | linkedin.com/in/gshrey | github.com/gshrey
Bengaluru, India

Summary

AI/ML engineer with production experience across revenue forecasting, fraud detection, and internal ML platforms. Works from GPU kernel optimisation and PyTorch training through to FastAPI services, agentic workflows, and deployed tooling.

Experience

• Agneyas Labs

Jul 2026 – Present

- Built and deployed an internal AutoML platform (Angular, Flask) that trains and benchmarks multiple algorithms per dataset, surfaces the strongest performer, and lets users run predictions and persist both models and results; in use by an internal team of 10–12.
- Developing a hybrid battery digital twin on proprietary EV cycling data: an equivalent-circuit model constrains known physics (terminal voltage, internal resistance, capacity fade) while a learned residual network absorbs nonlinear aging, cell-to-cell variance, and thermal effects.

• NetApp

Jun 2025 – May 2026

- Built and deployed a one-click forecasting pipeline for \$150M+ quarterly revenue planning using 1.5M+ records.
- Improved sales commission forecast accuracy by 15% with an XGBoost ensemble, reducing MAPE to 6–10% for GTM financial accruals across 4,900+ sales representatives.
- Designed scalable ML workflows for preprocessing, training, validation, and batch inference, reducing end-to-end runtime from 6–8 hours to 2 hours.

Projects

• Argus: Real-Time Fraud Detection Engine

[Code](#)

Tools: Python, XGBoost, Scikit-learn, SHAP, FastAPI, Next.js, Docker

- Built a real-time fraud scoring engine combining supervised XGBoost with an unsupervised Isolation Forest into one risk score, with per-transaction SHAP attributions in the analyst UI.
- Shared one causal feature function between training and serving, evaluated on a chronological holdout to prevent leakage, and streamed scores to a Next.js dashboard over WebSocket.

• RL Statistical Arbitrage Engine

[Code](#)

Tools: Python, Stable-Baselines3, Gymnasium, statsmodels, hmmlearn, MLflow

- Built regime-aware RL agents for cointegrated equity pairs: PPO over discrete entry/exit/flat actions and SAC over continuous position sizing, retrained per walk-forward fold under realistic transaction costs.
- Engineered a bias-controlled pipeline with Engle-Granger/Johansen tests, HMM regime features, point-in-time universe filtering, and MLflow-tracked runs; 94 tests including seed-determinism checks.

• SchemaBench: LLM Structured-Output Contract Benchmark

Tools: Python, JSON Schema, Pydantic, httpx, OpenRouter

- Built a benchmark measuring JSON Schema contract adherence across prompt-only, tool-call, and strict `response_format` modes, graded mechanically against the schema rather than by an LLM judge.
- Probe across 8 models found capability catalogues wrong in both directions: one advertising tool support returned HTTP 200 with an empty body, another advertising none honoured it; paired contrasts via exact McNemar, 175 tests.

• Inferbench: Triton Kernel for Batched LLM Decode

Tools: Python, Triton, CUDA, PyTorch, Transformers

- Diagnosed why batched fp16 decode ran *slower* than single-sequence on a tensor-core-less GPU — cuBLAS dispatches to a tensor-core GEMM at batch ≥ 2 — and fixed it with a Triton split-K kernel reaching $8.9\times$ on the projection and 52% of measured memory bandwidth.
- Raised end-to-end decode throughput $5.2\times$ on Qwen2.5-0.5B (87→451 tok/s at batch 16), gated on token-identical greedy output with negative controls at batch 32 and prefill where the kernel must do nothing; 182 tests.

Education

• Manipal Institute of Technology

Manipal, India

B.Tech in Computer Science and Engineering with Specialization in AI and ML | CGPA: 8.22

Sep 2022 – Jul 2026

Skills

- **Languages:** Python, SQL, C, C++, Java, Go, JavaScript, TypeScript, HTML, CSS
- **ML and AI:** PyTorch, Triton, CUDA, Scikit-learn, XGBoost, Stable-Baselines3, SHAP, FastAPI
- **Data Science:** Pandas, NumPy, Feature Engineering, Time-Series Forecasting, Reinforcement Learning
- **Tools and Platforms:** Docker, Git, Linux, MLflow, CI/CD, GitHub, Vercel, LeetCode